

DBMS

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WEEK-4: BOOKFLOW INTELLIGENT CATALOG & API BLUEPRINT

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-- 1. Create database

DROP DATABASE IF EXISTS bookflow_db;

CREATE DATABASE bookflow_db;

USE bookflow_db;

-- 2. Create Books table

CREATE TABLE Books

(

book_id INT AUTO_INCREMENT PRIMARY KEY,

title VARCHAR(200) NOT NULL,

author VARCHAR(150) NOT NULL,

genre VARCHAR(100),

description TEXT,

published_year INT

);

-- 3. Insert 10 books

INSERT INTO Books

(

title,

author,

genre,

description,

published_year

)

VALUES

(

'The Martian',

'Andy Weir',

'Science Fiction',

'An astronaut struggles to survive alone on Mars.',

2011

),

(

'Dune',

'Frank Herbert',

'Science Fiction',

'A story involving planets, politics, survival and space.',
1965
)
(
'The Hitchhikers Guide to the Galaxy',
'Douglas Adams',
'Science Fiction',
'A humorous adventure across space and alien worlds.',
1979
)
(
'Pride and Prejudice',
'Jane Austen',
'Romance',
'A classic story about relationships and social expectations.',
1813
)
(
'The Great Gatsby',
'F. Scott Fitzgerald',
'Classic',
'A story of ambition, wealth, love and the American dream.',
1925
)
(
'Foundation',
'Isaac Asimov',
'Science Fiction',
'A futuristic story about mathematics, civilization and space.',
1951
)
(
'The Hobbit',
'J.R.R. Tolkien',
'Fantasy',
'A fantasy adventure involving Bilbo Baggins and a dangerous journey.',
1937
)
(
'Enders Game',
'Orson Scott Card',
'Science Fiction',
'A young student is trained for an interstellar conflict.',
1985

```

),
(
  'Harry Potter and the Sorcerers Stone',
  'J.K. Rowling',
  'Fantasy',
  'A young wizard begins his magical education.',
  1997
),
(
  'Twenty Thousand Leagues Under the Seas',
  'Jules Verne',
  'Adventure',
  'A fantastic underwater adventure aboard a mysterious submarine.',
  1870
);

```

```

-- 4. Add embedding column
ALTER TABLE Books
ADD COLUMN embedding JSON;

```

```

-- 5. Insert mock vectors

```

```

UPDATE Books
SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
WHERE book_id = 1;

```

```

UPDATE Books
SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
WHERE book_id = 2;

```

```

UPDATE Books
SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
WHERE book_id = 3;

```

```

UPDATE Books
SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
WHERE book_id = 4;

```

```

UPDATE Books
SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
WHERE book_id = 5;

```

```

UPDATE Books
SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)

```

```
WHERE book_id = 6;
```

```
UPDATE Books  
SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)  
WHERE book_id = 7;
```

```
UPDATE Books  
SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)  
WHERE book_id = 8;
```

```
UPDATE Books  
SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)  
WHERE book_id = 9;
```

```
UPDATE Books  
SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)  
WHERE book_id = 10;
```

```
-- 6. Check all books and vectors
```

```
SELECT  
    book_id,  
    title,  
    author,  
    genre,  
    published_year,  
    embedding  
FROM Books  
ORDER BY book_id;
```

Result Grid						
Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:
book_id	title	author	genre	published_year	embedding	
1	The Martian	Andy Weir	Science Fiction	2011	[0.90, 0.80, 0.20]	
2	Dune	Frank Herbert	Science Fiction	1965	[0.85, 0.75, 0.25]	
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979	[0.80, 0.70, 0.30]	
4	Pride and Prejudice	Jane Austen	Romance	1813	[0.10, 0.20, 0.90]	
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925	[0.15, 0.30, 0.80]	
6	Foundation	Isaac Asimov	Science Fiction	1951	[0.95, 0.65, 0.20]	
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937	[0.30, 0.75, 0.90]	
8	Enders Game	Orson Scott Card	Science Fiction	1985	[0.88, 0.85, 0.15]	
9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997	[0.20, 0.55, 0.95]	
10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870	[0.70, 0.90, 0.40]	
NULL	NULL	NULL	NULL	NULL	NULL	

```
SELECT  
    book_id,  
    title,
```

```

author,
genre,
ROUND(
(
  CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) * 0.90
  +
  CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) * 0.80
  +
  CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) * 0.20
)
/
(
  SQRT(
    POW(CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)), 2)
    +
    POW(CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)), 2)
    +
    POW(CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)), 2)
  )
  *
  SQRT(
    POW(0.90, 2)
    +
    POW(0.80, 2)
    +
    POW(0.20, 2)
  )
),
4
) AS cosine_similarity
FROM Books
WHERE embedding IS NOT NULL
ORDER BY cosine_similarity DESC
LIMIT 3;

```

Result Grid					
		Filter Rows:			Export:
				Wrap Cell Content:	Fetch rows:
	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983